HUMANITARIAN ENGINEERING: POLITICS AND PRACTICE – A NEW INTERDISCIPLINARY COURSE ON THE APPLICATION OF ENGINEERING SKILLS TO HUMANITARIAN CHALLENGES

Jenny Peterson[1] and Gabriel Potvin[2]
[1] Department of Political Science, University of British Columbia
[2] Department of Chemical and Biological Engineering, University of British Columbia
jenny.peterson@ubc.ca; gabriel.potvin@ubc.ca

Abstract – *Background and Purpose: Engineers have an important role to play in the tackling of the world's complex sustainability and social challenges, solutions to which must, by necessity, involve both technical and social components. As engineers require increasing sociotechnical and interdisciplinary fluency, so too do our curricula need to change. Approach: A new course, Humanitarian Engineering: Politics and Practice, was created in which engineering and arts students work in mixed-discipline teams to propose solutions to real-world humanitarian challenges provided by partnering NGOs. The course introduces engineering design, humanitarian theory, and critical political analysis through interdisciplinary lenses. Outcomes: Survey responses suggest that this course provided a unique and rewarding opportunity for students that helped develop their skills in interdisciplinary collaboration, and shaped the way they think about their professional roles. Conclusions: Given the success of this first offering of the course, it will be offered again, alongside a more deliberate study of learning outcomes of this interdisciplinary approach and its scalability.*

Keywords: Interdisciplinary Education, Humanitarian Engineering, Engineering Design, Experiential Learning

1. INTRODUCTION

Engineers have an important role to play in addressing the daunting sustainability challenges the world is facing, including climate change, public health crises, and social inequality. Effective solutions to these complex challenges must, by necessity, integrate both social and technical considerations to be successful. Engineering curricula, however, typically focus almost exclusively on the latter, with social considerations usually covered as core content in one or two dedicated courses, likely undercutting their importance with students [1]-[3]. If engineers are to play a greater leadership role in tackling these challenges, a proposition typically supported by and advocated for by engineering programs and instructors, they will require more sociotechnical and interdisciplinary fluency, which highlights an increasingly conspicuous lacuna in our engineering curricula [4]. At the same time, many students report studying engineering to "have a positive impact on the world" but feel their programs do not offer them sufficient opportunities to pursue these objectives [5], [6].

Though strong technical proficiency and design skills remain the core skillset of engineers, there is a growing recognition of the limitations of traditional disciplinary silos in addressing multifaceted challenges such as the ones listed above [4]. Interdisciplinary collaboration has emerged as a powerful tool for addressing such challenges by leveraging diverse perspectives, methodologies, and expertise. Interdisciplinary collaborative problem-based pedagogical approaches have also been shown to support the development of transferrable skills including critical thinking, communication, problem-solving, and teamwork [7]-[9], all of which remain core desired outcomes of engineering curricula.

Given the above and inspired by a student-led seminar course described in previous work [10], a new course, *Humanitarian Engineering: Politics and Practice*, was created at the University of British Columbia (UBC) and delivered for the first time in the fall of 2023. In this course, engineering, political science, and international relations students collaborate and apply their training to address real-world humanitarian challenges provided by partnering NGOs. This course, cross-listed in the Faculties of Applied Science and Arts at UBC and co-taught by engineering and political science professors (the authors of this paper), was designed to broaden students' appreciation of the roles and contributions that the varied disciplines have in tackling complex problems, and to provide authentic experience in interdisciplinary collaboration and teamwork, all while allowing students to have a real, tangible impact by applying their expertise in the service of partnering NGOs. Whether or not students continue working in the humanitarian aid field after graduation, the skills honed in this course are highly transferrable to any industrial setting

This work is licensed under a Creative Commons Attribution-NonCommercial 4.0 International License.

and in high demand by engineering employers. Though there has been a worldwide increase in "Humanitarian Engineering" programs, courses, and training certificates over the last few years [11], this course is unique in its interdisciplinarity, in terms of the student participants, pedagogical and instructional approaches and practices, course deliverables and assessments, and experiential learning opportunities.

The inaugural cohort of the course was composed of 26 engineering students, from nine different disciplines, and 28 arts (political science or international relations) students, working on nine unique projects provided by seven partnering NGOs. This paper presents the structure of the course, the pedagogical and instructional strategies used in its design, describes its key deliverables, and summarizes student impressions of the course collected through midterm and end-of-term surveys, and instructor reflections on this interdisciplinary collaboration and its outcomes. Key challenges and opportunities for improvement are also highlighted, ahead of its second iteration during the 2024-2025 academic year. The objective is to outline the core structure of this new course and highlight the potential of this approach in training engineers, and set the stage for future, more focused studies on interdisciplinary learning outcomes.

2. COURSE DESCRIPTION

The following outlines the course structure, content, assessments, and pedagogical approaches used in the design and delivery of this course. Only a brief overview is presented here, but the authors are happy to share more detailed material with interested readers.

2.1. Course Structure and Content

The course (cross-listed as APSC 367 and POLI 371 at UBC) meets once a week for a three-hour session. The first half of the session was allocated to exploring content through lectures or interactive workshops, with the remaining time in the session dedicated to teams working together on the term project (discussed in section 2.3 below). The first lecture in the course was dedicated to discussions on interdisciplinarity and interdisciplinary teamwork, as well as exploring preconceived ideas on the professional identities and values of engineers and social scientists, and strategies to facilitate interdisciplinary dialogue, all to generate the buy-in necessary for the course to proceed smoothly. In subsequent weeks, sessions focused on topics relevant to their term projects, including an introduction to the engineering design process, humanitarian theory, critical context and political analysis frameworks, sustainability and sustainable design, and project risk assessment, implementation and monitoring, all through interdisciplinary lenses. Three sessions were dedicated to "technical modules" on renewable energy, water treatment, and genetic engineering for food production, which presented technical scientific and engineering concepts to establish a link between the higher-level design decisions made by teams and the actual implementation of their proposed solutions, and gain an appreciation for the technical and logistical challenges inherent to many humanitarian projects. Humanitarian workers, technical engineers and industry leaders were brought in as guest speakers to supplement the content where appropriate.

This course was approved to count as the "Impact of Engineering on Society, Sustainability and Environmental Stewardship" requirement that all engineering students at UBC have to meet to graduate. It also counted for credit towards political science and international relations degrees. It should be noted that students had to apply to register in this course by completing a short survey in which they had to present their motivation for taking this course. This ensured that enrolled students were all committed to the undertaking, and allowed the number of students from engineering and arts to remain approximately equal, which was a desired feature of the course.

2.2. Instructional Strategy

One of the key features of this course is that the content is fully integrated, and that each lecture is co-taught by both course instructors (i.e. there are no "engineering" lectures or "political science" lectures, and all content is covered jointly through an interdisciplinary lens). This allows instructors to share their expertise and perspectives, while reinforcing or challenging each others' approaches or positions in open conversation, in order to model productive interdisciplinary dialogue. As evidenced by student feedback, this feature of the course was particularly successful and seen as valuable.

2.3. Term Project

Students work in mixed-discipline teams (two or three engineering and two or three political science/international relations students in each team) on one of nine real world-spanning design projects, provided by seven partnering NGOs, aiming to tackle real humanitarian challenges using a designed technological intervention. Throughout the term, each team remotely consulted with a representative of their partnering organization once a month as they worked on different aspects of the project, to exchange ideas and receive feedback on their progress. At the end of the term, student teams delivered comprehensive project reports to their NGOs, which included detailed political and context analyses, technical designs and schematics for their recommended approaches, holistic sustainability assessments, technical, social, and economic risk analyses, and project implementation, evaluation, and monitoring plans for their proposed solutions. An IP transfer agreement was signed by all participants to allow the NGOs to use the content of the report in their work upon

its completion. Students presented a preliminary project proposal as a midterm oral presentation, and then showcased their final projects as posters during an end-of-term reception. To encourage genuine interdisciplinary collaboration and exchange, as opposed to a divide-and-conquer distribution of tasks, during oral presentations the engineering students were required to present the social and political analysis components of the project, and the arts students had to present and explain the technical design-related aspects of their proposal. The project represented 50% of the final grade of the course, distributed among the different deliverables.

2.4. Case Studies and Reflections

In addition to the project, students had to complete three individual assessments. The first two were case studies in which students critically assessed an engineering failure and an engineering success in humanitarian aid respectively, drawing from social and political analysis frameworks discussed in class. The final assessment was a self-reflection on students' professional identities, their experience in the course, and their thoughts on the role of their discipline in humanitarian aid. Collectively these deliverables were worth 30% of each student's final grade, with the remaining grade distribution allocated to participation in all aspects of the course.

3. COURSE EVALUATION METHODS

Two surveys were sent to students during the course, one at the mid-point of the term, and one at its conclusion. Participation in these surveys was voluntary, and responses were anonymous, though students were asked to identify whether they were an engineering student or an arts student. The purpose of the surveys was explicitly for course feedback, in the context of this being its inaugural offering. Participants were informed that the answers and feedback provided were to be used for course review and improvement, and that they would be shared with other instructors and the broader education community through publications.

The first survey was sent to students at the mid-point of the term, immediately after the midterm project progress reports and presentations were completed. It consisted of the thirteen, five-point Likert-scale questions listed below assessing participants level of agreement with the provided statements, the first seven of which focused on students' general experiences in the course, and the last six of which focused on students' experience with the project as the course's main deliverable.

1. *I find the content of the course interesting.*
2. *I have learned/am learning a lot in this course.*
3. *This course aligns well with the expectations I had when I signed up for it.*
4. *I feel the course content is relevant to humanitarian work.*
5. *Interdisciplinary is important in education and my future career.*
6. *Opportunities for interdisciplinary collaboration are effectively integrated in this course.*
7. *Having both instructors (engineering and political science) co-teaching each lecture is a good thing in this course.*
8. *Working in interdisciplinary teams is an important feature of this course.*
9. *Working in my team has been a positive experience so far.*
10. *The format and content of the project are appropriate for this course.*
11. *Working with NGO-supplied projects is a positive aspect of this course.*
12. *Meeting with my NGO has been a positive experience.*
13. *I feel suitably equipped to tackle the challenges presented by this project.*

Students also had the opportunity to provide feedback or explain their answers in an open-ended text box at the end. 15/54 students participated, with approximately equal participation from both student groups (7/26 engineering students, and 8/28 arts students).

The final survey was sent to students at the conclusion of the course, immediately after the final projects were submitted and after the project poster showcase. It consisted of six narrative-driven open-ended question prompts that built on themes identified in the responses to the midterm survey, and aimed to gain additional insight into the students' experience with interdisciplinarity in general and with the term project. The six prompts were:

i. *You worked on interdisciplinary content (lectures, readings) and assignments (case studies) all term. Regardless of your program of study, this included engagement with material outside of the core content in your own discipline. What, if anything, did you find particularly challenging about this? Are there additional resources that would have helped you better or more easily engage with the material?*
ii. *Part of the purpose of the course was to gain an appreciation for what the "other disciplines" do, as well as gain an appreciation for what your own discipline can contribute to real challenges. Do you feel this has been successful? Do you better understand and/or find more value in what other roles contribute to difficult problems like humanitarian challenges? Has your own disciplinary identity or your view of your own discipline's roles changed after taking this course? How so?*
iii. *What were the good aspects of the project? What went well or what did you enjoy? What should be kept for the*

next iteration of the course? This could relate to the project structure and format itself, or your own experience working on it, both individually or as part of your (mixed-disciplines) group.

iv. *What were the challenging aspects of the project? What could have gone better or what did you dislike? What should be changed in the next iteration of the course? This could relate to the project structure and format itself, or your own experience working on it, both individually or as part of the (mixed-disciplines) group.*

v. *Some aspects of the project required you to work outside of your discipline and to do a lot of research or develop new skills on your own. Do you feel you were suitably equipped for this and able to do it? If not, what type of extra training would have been useful to better equip you to tackle this project?*

vi. *You can add any other comment or feedback about the course you want here.*

Only 9/54 students responded to this survey, with the drop in participation attributed to the timing of its release (i.e. after the course was over). Responses were thematically analyzed by both instructors.

It should be noted that neither of these surveys was based on a validated survey instrument, and were meant to collect targeted feedback on the course structure and delivery, and experience working on its unique deliverables, during its first offering. Future iterations of the course may lend themselves to more comprehensive study of these topics.

4. RESULTS AND DISCUSSION

The following outlines the results of both surveys, along with instructor reflections or observations about the course. Please note that results are only summarized, and that detailed results are only shown to highlight points of particular interest. Excerpts of student responses are provided in italics to support the points being discussed or to provide context as necessary. The authors are, however, more than happy to discuss the results further with interested parties. It should be noted that the number of survey participants, as stated in section 3, is relatively low, though in-line with expectations given the voluntary nature of participation, and the timing of survey deployment. The representativeness of results should therefore be interpreted in this context.

4.1. Midterm Survey Results

A summary of scores for each question, broken down by discipline, is shown in Table 1 below.

The following outlines key results from the midterm survey:

i. At the midpoint of the course, students found the course interesting (14/15 of the responses to Q1 were positive, defined as responses of 'Strongly Agree' or 'Somewhat Agree', with the remaining response being neutral), relevant to humanitarian work (15/15 responses to Q4 were positive), and relevant to their careers (14/15 Q5 responses positive, and 1/15 neutral).

Table 1. Summary of responses to midterm survey

Course-Related Questions				Project-Related Questions			
		Average Score*	Mode			Average Score*	Mode
Q1	**Overall**	4.38	4	**Q8**	**Overall**	4.44	5
	ENG	4.57	5		**ENG**	5.00	5
	POLI/IR	4.25	4		**POLI/IR**	4.50	5
Q2	**Overall**	4.13	5	**Q9**	**Overall**	3.81	5
	ENG	3.86	5		**ENG**	4.14	5
	POLI/IR	4.38	4		**POLI/IR**	4.00	5
Q3	**Overall**	3.88	5	**Q10**	**Overall**	4.25	5
	ENG	4.00	5		**ENG**	4.43	5
	POLI/IR	3.88	5		**POLI/IR**	4.63	5
Q4	**Overall**	4.63	5	**Q11**	**Overall**	4.38	5
	ENG	4.86	5		**ENG**	4.71	5
	POLI/IR	4.50	5		**POLI/IR**	4.63	5
Q5	**Overall**	4.75	5	**Q12**	**Overall**	4.13	5
	ENG	5.00	5		**ENG**	4.29	5
	POLI/IR	4.50	5		**POLI/IR**	4.50	5
Q6	**Overall**	4.31	5	**Q13**	**Overall**	3.63	4
	ENG	4.43	5		**ENG**	3.86	4
	POLI/IR	4.75	5		**POLI/IR**	3.88	4
Q7	**Overall**	4.94	5				
	ENG	5.00	5				
	POLI/IR	4.88	5				

* 1 - Strongly Disagree, 2 - Somewhat Disagree, 3 - Neutral, 4 - Somewhat agree, 5 - Strongly agree

ii. The majority of respondents (13/15) reported learning a lot in this course in Q2. Interestingly, the two negative responses (both 'Somewhat Disagree') were engineering students, with all other engineering students selecting 'Strongly Agree' for this statement.

All 'Somewhat Agree' responses were from Arts students. Both of the engineering students responding negatively to the prompt indicated difficulty applying political science concepts to the case studies.

"I don't understand so many of the "isms", schools of thought, and meanings of other jargon that all the poli sci students seem to have a background in. For a

similar reason, the case study was quite difficult to research and write since I didn't have many concepts to apply."

This also aligns well with the Q6 results, on the effectiveness of integrating interdisciplinarity, to which the same 2 engineering students also answered negatively, with the rest of the respondents responding very positively to the interdisciplinary approaches used.

iii. Responses to Q7 confirm the perceived value of having both instructors teach every lecture, with all responses being positive.

iv. The overall impression of the project and was quite positive, as evidenced by the answers to Q8 and Q10 (14/15 positive responses, 1 neutral response). Similarly, experiences working with partnering NGOs was positive, as evidenced by the responses to Q11 and Q12.

The most polarized student responses were obtained for Q13, which asked students if they felt suitably equipped to tackle the term project. The results are shown in Figure 1. Although results suggest that the majority of students do feel able to tackle the challenges presented, a good proportion of participating students may not feel fully equipped to participate in this interdisciplinary project. This seems to include students from both engineering and arts.

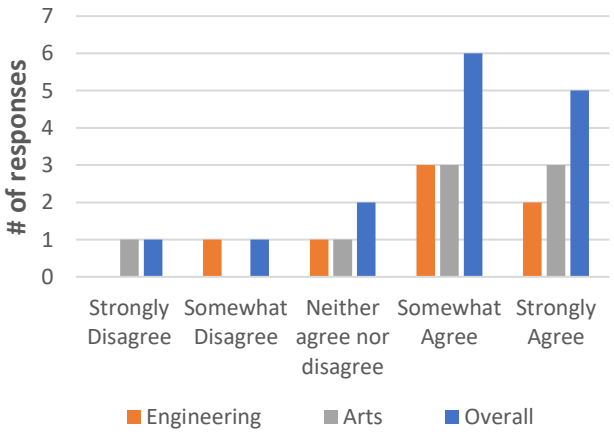


Fig. 1. Student responses (n = 15) to Q13: I feel suitably equipped to tackle the challenges presented by the project.

Responses to the open-ended question provide additional context, and suggest two explanations for this reported lack of confidence. The first, in line with the comment about the course presented above, consists of a feeling of not being able to, or being reluctant to, contribute to the discussions outside of one's own expertise.

Arts Student: *"I do, as a POLI student feel slightly unequipped to tackle some project challenges, as the bulk of our project seems to be the technical side."*

Engineering Student: *"I feel mostly suitably equipped to deal with the engineering side of things, but I wish I had more background on the social sciences in order to unlock more insight and perspectives that are relevant to humanitarian work that are not taught in engineering classes, or even in other places outside of the social sciences themselves."*

The second concerns interaction with partnering NGOs. Some students reported that their NGOs did not define their problem statement well, or did not provide sufficient information to allow them to feel confident in proposing a suitable solution. Others reported that the NGOs already had a good grasp of the problem and a suitable solution, and did not feel confident that they would be able to further contribute meaningfully to its resolution.

"Not having all the information at hand that you would want, makes things difficult because you want to do meaningful work but you also don't want to disturb your NGO with every little detail. I am just not looking forward to making a LOT of assumptions but I don't really know how you would be able to change that for a project like this."

It is interesting to consider that these responses were provided by students after their midterm presentations, in which each team did, in fact, produce high quality and valuable work, belying their concerns about their abilities. This suggests that the course may benefit from additional framing of interdisciplinary collaborations and individuals' roles in the group, as well as a better primer of what to expect when dealing with humanitarian aid organizations, which often have limited information about the details of projects at their inception. This can be refined in subsequent iterations of the course. Though students were concerned about working with limited or incomplete information, and having to rely on assumptions or estimates to complete their work, this is part of the reality of working on these types of real-world projects, and this may therefore highlight a gap in the training provided in our overall curricula. Additional support and expectation-setting will be considered in the next iteration of the course to address this.

4.2. Final Survey Results

The limited number of responses to the final survey (n = 9/54) does not lend itself well to the drawing of strong conclusions, and a full thematic analysis of these responses

is therefore not presented here. There were, however, three themes related to course challenges that are worth highlighting here.

i. Logistical Challenges: Multiple comments highlighted that the format of the course (i.e. a single 3-hour session once a week for a term) was not ideal, and preference for dividing these sessions into two shorter session, one of which should still be allotted to group work on the term project, with the dedicated teamwork time, with instructor support, being valued. There was also a general sense that given the scope of the project, a single term was not enough to produce a comprehensive proposal for their assigned challenge, and moving to a two-term course was recommended.

ii. Better integration of the technical modules: The three technical modules were seen by some as "separate" from the course's flow and not directly relevant to the project or the overall curriculum or narrative. Strategies for better integration of this content will be examined for the next iteration of the course.

iii. Better screening of NGO partners and projects: Perhaps unsurprisingly given the number of partnering NGOs, the projects assigned to teams came with different levels of detail, information, and supporting information. Several teams were frustrated at having to make significant assumptions that they felt undermined their ability to provide a relevant proposal to address the challenges, compounded by the fact that other teams had much more comprehensive data sets and more informed NGO partners.

"[The problem] was getting usable information from the NGO, mostly (engineering related) data. […] I was just really disappointed because I believe that we could have done a way better job with our project had we had the information from the beginning. What I'm taking from this or what I would suggest for the next iteration of this course, would be to kindly ask the NGOs to have some specific data ready from the start."

All of these concerns can be addressed in the next iteration of the course, as discussed above.

The most common themes in the responses were very positive, and highlighted the value of interdisciplinary collaboration for tackling these challenges, with such responses coming from both engineering and arts students. One response is highlighted below. A more comprehensive and deliberate investigation of learning outcomes and the effect of this course on disciplinary identity and perceptions of professional roles will be conducted in the future (i.e. in the next iterations of the course), but these responses suggest that this type of collaboration can help broaden and contextualize professional roles, and highlight the need for lifelong learning, all of which align well with curricular objectives in engineering.

"My view on my own [engineering] discipline definitely changed. […] I now think that the basics of social science should be second nature, because if I am to work in a field like humanitarian engineering, I must internalise that there is no scenario where the technology I use exists isolated from people. I do think that the arts people offered POVs that I could never have considered by myself […] while I was stuck with my own basic POV on the subject."

4.3. Instructor Reflections

The interdisciplinarity of this course proved fruitful and rewarding not just for the students, but for the instructors as well. Co-designing and co-teaching each lecture provided insight and appreciation for the way each discipline approached problems, and though it required significant effort, compromise, and navigating philosophical and pedagogical disagreements, it also facilitated an exchange of best practices, and a reframing of disciplinary identities within academia, and ultimately highlighted common goals and values. We both feel as if we leave this first iteration of the course as better educators and better disciplinary practitioners. Modeling interdisciplinary dialogue for the class by openly questioning or challenging approaches or perspectives presented by the other instructor, though seen as unusual by some respondents, emerged as an invaluable pedagogical tool, as reflected in survey responses.

"Also it's fantastic having two instructors from different disciplines. Hearing their different approaches on challenges and problems in the world of humanitarian engineering is a great thing to experience, especially because they don't necessarily have the same opinion on everything but they respect each others point of view and discuss them partly in front of class. That way as a student, it gets you thinking because you directly see that there is more to it than one single approach. That's very different to a lot of courses where you usually only have one instructor with his/her viewpoint."

As we reflected on the course, another point that struck us is that as the term progressed, and as we interacted with the teams working on their projects, it was increasingly difficult to identify which students were from engineering, and which were from political science or international relations. Of course, as expected and intended, students took the lead on project sections most aligned with their expertise, and the different components of the final deliverables demonstrated strong disciplinary skillsets, but it was clear that all team members were actively communicating and exchanging ideas, and that each participant was able to discuss all aspects of their project

on their own. This was echoed by invited guests during the final poster presentation, who were surprised by this. It should of course be noted that students had to apply to take this course, which likely self-selected for those students that were interested in interdisciplinary collaboration. Even taking this into account however, this observation suggests that the often-perceived barriers in disciplinary skillsets may not be as immovable as we think, and that given proper support, structure, and motivation, there may be opportunities for more value-added interdisciplinary education in different contexts in engineering programs.

4.4. Anecdotal Results

Finally, there are two anecdotal but interesting observations that are worth noting here.

The first is that the regularly-administered student evaluations of teaching, aggregated separately for engineering and arts students, yielded identical average scores from both groups. Though the limits of these evaluations in assessing the quality of instruction are well documented, the parity of scores suggests that the course content was equally well received by all participating students, which speaks to the success of the interdisciplinary approach used, at least in terms of student engagement and perceived value of the content.

The second point of interest is that over the course of the term, two of the engineering students enrolled in the course privately and independently effusively expressed to us (the instructors) that this opportunity to apply their engineering skills to these real projects and have an impact restored their enthusiasm for their disciplines, and convinced them not to drop out of the program as they were planning on doing. Though, obviously, no wider conclusions can be drawn from this, it does serve as an indication that this type of course may fulfill the desire to have an impact felt by a subset of engineering students.

5. Conclusions and Future Work

There are very few opportunities in traditional engineering curricula for cross-faculty or interdisciplinary collaborations. Though this interdisciplinary course was only offered for the first time, the quality of the work produced, which would not have been completed at this caliber in single-discipline teams, student feedback, and instructor observations all suggest that this kind of approach, though time- and effort-intensive, presents excellent opportunities for the development of transferrable skills in engineering students, and student engagement through authentic, real-world design work. The skills in interdisciplinary teamwork and fluency, the development of which is increasingly valued in engineering programs, are facilitated by creating such opportunities.

The goal of this paper was to introduce this course and perhaps highlight the potential of this approach in terms of modernizing the training of engineering students. In addition to implementing a few changes in the next iterations of the course, a more detailed and deliberate set of studies will be developed to take a closer look as engineering skill development, learning outcomes, and sense of disciplinary identity and professional belonging in participants. Given the level of interest for participation by both students and NGOs, opportunities to scale this course will also be investigated.

Acknowledgements

Thanks are extended to EcoSwell, Energy for All, Mercy Corps, RES'EAU, CAWST, the Sehgal Foundation, and Get Water Uganda for participating as the inaugural NGO partners in this course. Thanks are also extended to the Faculties of Applied Science and Arts at UBC for their support of this unique course.

As established by the UBC Behavioural Research Ethics Board (BREB) office, the work is considered "Quality Improvement and Assurance and Program Evaluation", which under Article 2.5 of the Tri Council Policy Statement is exempt from institutional ethics review requirements. This work therefore does not require ethics review for its performance or dissemination.

References

[1] A. Bielefeldt and N. Canney, "Changes in the Social Responsibility Attitudes of Engineering Students Over Time", in *Sci. Eng. Ethics*, 2016 (22), pp. 1535-1552. https://doi.org/10.1007/S11948-015-9706-5

[2] J. Lim, N. Findlater, C. Interiano, P. Tkacik, J. Dalhberg, L. Faw, B. Hunt, and N. Nguyen, ""Our Little World": (Un)envisioning the Social and Ethical Implications of Engineering" in *2018 IEEE Frontiers in Ed. Conf. (FIE)*, 2018, 1-6. https://doi.org/10.1109/FIE.2018.8659303

[3] S. Niles, S. Contreras, S. Roudbari, J. Kaminsky, and J. Harrison, "Resisting and assisting engagement with public welfare in engineering education" in *J. of Eng. Ed.* 2019, 109, 491 - 507. https://doi.org/10.1002/jee.20323

[4] J.A. Leydens and J.C. Lucena. "Transforming Engineering Education and Practice " in *Engineering Justice: Transforming Engineering Education and Practice*. Hoboken, New Jersey, John Wiley & Sons, 2018.

[5] P. Jiménez, J. Pascual, and A. Mejía, "Towards a pedagogical model of social justice in engineering education" in *Proc. 8th Int. Conf. on Software and Information Eng*. (2019) https://doi.org/10.1145/3328833.3328861

[6] G. Rulifson, and A. Bielefeldt, "Motivations to Leave Engineering: Through a Lens of Social Responsibility" in *Eng.*

[7] A. Ana, L. Alhapip, M. Barliana, Y. Rahmawati, M. Anon, and V. Dwiyanti, "Transferrable

Skills Needed in the Workplace", in *J. Eng. Ed. Transformations*, 2020. https://doi.org/10.16920/JEET/2020/V34I0/157858

[8] R. Collins-Nelsen, F. Koziarz, B. Levinson, E. Allard, S. Verkoeyen, and S. Raha, "Social context and transferable skill development in experiential learning" in *Innovations in Ed. and Teaching Int.*, 2021, 59, pp. 421 - 430. https://doi.org/10.1080/14703297.2021.1886970

[9] J. Mitchell, A. Nyamapfene, K. Roach, and E. Tilley, "Philosophies and pedagogies that shape an integrated engineering programme" in *Higher Education Pedagogies*, 2019, 4, 180 - 196. https://doi.org/10.1080/23752696.2018.1507624

Studies, 2017, 9, pp. 222 - 248. https://doi.org/10.1080/19378629.2017.1397159

[10] A. Sheroubi, and G. Potvin, "Humanitarian Engineering: A New Interdisciplinary Course on the Application of Engineering Skills to Local and Global Humanitarian Challenges." in *Proc. 2018 Can. Eng. Ed. Assoc. Conf*. 2018, Paper 8.

[11] J. Smith, A. Tran, and P. Compston, "Review of humanitarian action and development engineering education programmes" in *European J. Eng. Ed.*, 2020, 45, pp. 249 – 272, https://doi.org/10.1080/03043797.2019.1623179